# Experiment/Practical 12 Project /Case Study on Classification

**Vedant Chavan**
**25070126218**
**AIML C1**

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## 1. Title:

Monkeypox Detection from Skin Images Using Deep Ensemble Learning and Explainable AI: A Comparative Study of Xception, DenseNet201, and EfficientNetB7 with Grad-CAM Interpretability

## 2. Aim:

To develop and evaluate a deep learning-based ensemble classifier for binary detection of Monkeypox from skin lesion images using transfer learning with three state-of-the-art CNN architectures (Xception, DenseNet201, EfficientNetB7), and to provide visual explainability using Grad-CAM to validate the clinical relevance of model predictions, in alignment with recent literature on Mpox detection.

## 3. Problem Statement

Monkeypox (Mpox) is a viral zoonotic disease that re-emerged as a global public health emergency in 2022, with over 89,000 confirmed cases reported across 113 countries by August 2023 [1]. The disease manifests with skin lesions that closely resemble chickenpox, measles, and other dermatological conditions, making early and accurate diagnosis challenging. Although PCR-based testing is reliable, it requires specialized infrastructure, trained personnel, and cold-chain logistics that are often unavailable in resource-constrained settings [2]. Recent research has demonstrated that deep learning models applied to skin lesion images can achieve high accuracy in distinguishing Mpox from similar conditions [3, 4]. However, most existing studies rely on a single model architecture, which is prone to overfitting, class imbalance issues, and poor generalization [5]. Furthermore, few studies integrate explainability — clinicians cannot interpret why a prediction was made, which limits clinical trust and adoption [4]. This project addresses these gaps by (a) comparing three state-of-the-art CNN architectures, (b) combining them into a soft-voting ensemble to improve robustness, and (c) applying Grad-CAM to visualize which regions of the image drive predictions, following the methodology established in Mpox-XDE [1] and B2MVIT [2].

## 4. Literature Work *(Minimum 5 recent papers from good quality journal):*

*Table 1: Summary of recent literature on Mpox detection*

| # | Author(s) / Year | Title & Journal | Dataset | Method | Accuracy | XAI |
|---|---|---|---|---|---|---|
| 1 | Saha, D.K., et al. (2025) | Mpox-XDE: an ensemble model utilizing deep CNN and explainable AI for monkeypox detection and classification. BMC Infectious Diseases | MSID (770 images, 4 classes) | Ensemble (Xception + DenseNet201 + EfficientNetB7) with Softmax, dense, flatten, 65% dropout | 98.70% | Grad-CAM |
| 2 | Agrawal, T., et al. (2026) | A Robust Transformer Architecture with Broad Attention for Monkeypox Lesion Detection in Skin Images. Int. J. of Computational Intelligence Systems (Springer) | MSLD (3,192 augmented) & MSID (23,100 augmented) | B2MVIT (ViT backbone + Broad Attention + Mini-MVIT encoder + feature fusion) | 99.69% (MSLD), 99.19% (MSID) | Grad-CAM |
| 3 | Hossain, M.S., et al. (2025) | From survey to solution: A deep learning framework for reliable monkeypox diagnosis using skin images. Array (Elsevier) | MPox-Vision (800 images, 4 classes) + MSID, MCSI, MSLDv2.0 | Two-stage hybrid ensemble: top-4 deep models + KNN meta-classifier | 98.70% | SHAP |
| 4 | Mustahid, A.A.M., et al. (2026) | MicroMPoxNet: A lightweight CNN for interpretable Monkeypox classification in dermoscopic images with comparative transfer learning evaluation. Biomedical Signal Processing and Control (Elsevier) | Mpox2025 (696 images) + public dataset (6 classes) | Lightweight CNN (SCRAN + SE blocks), 0.63M parameters | 99.99% (binary), 99.12% (multiclass) | Grad-CAM, LIME, IG |
| 5 | Saha, D.K., et al. (2025) | Mpox-XDE (dataset and preprocessing reference). BMC Infectious Diseases | MSID (770 images) | Modified Xception, DenseNet201, EfficientNetB7 with 224×224 preprocessing | — | — |

*Table 2: Method, performance and research gap of the reviewed papers*

| Reference | Method | Performance | Research Gap |
|---|---|---|---|
| Saha et al. (2025) — Mpox-XDE [1] | Ensemble of 3 modified DL models (Xception + DenseNet201 + EfficientNetB7) combined via Softmax layer, dense layer (1024 neurons), flatten layer, 65% dropout and global average pooling; Grad-CAM for XAI; MSID (770 images, 4 classes). | Accuracy 98.70%, Precision 98.90%, Recall 98.80%, F1 98.80% | (1) Small dataset (770 images) may cause overfitting and poor generalization to real clinical data.<br>(2) Complex ensemble increases training time (4.86 h) and reduces per-layer interpretability.<br>(3) Binary or 4-class only; cannot separate Mpox from mimics such as cowpox/smallpox.<br>(4) High RAM (9.5 GB) and GPU (14.4 GB) usage; not deployable on edge devices.<br>(5) No statistical validation (no Chi-Square or paired t-test). |
| Agrawal et al. (2026) — B2MVIT [2] | B2MVIT-Net: ViT backbone (12 encoder layers) + Broad Attention + Mini-MVIT encoder + deep feature extraction + feature fusion ($\gamma = 0.5$); Grad-CAM; MSLD (3,192 augmented) & MSID (23,100 augmented); 5-fold CV + Chi-Square + paired t-test. | MSLD 99.69%, MSID 99.19%; Precision/Recall/F1 ≈ 99%; AUC = 1.00; Chi-square = 621.86 ($p < 0.05$) | (1) Binary only (Monkeypox vs Non-Monkeypox); cannot differentiate from chickenpox or measles.<br>(2) High computational cost: 85.65M parameters, 16.86 GFLOPs, 20.3 ms inference; unsuitable for real-time edge use.<br>(3) Not validated on multi-centre or real-time clinical data.<br>(4) Dataset bias: standardized images lack diversity of skin tones, lighting, lesion types and devices.<br>(5) Relies heavily on augmented images that may not reflect real clinical variability. |

| Reference | Method | Performance | Research Gap |
|---|---|---|---|
| Hossain et al. (2025) — From Survey to Solution [3] | Two-stage hybrid ensemble: top-4 deep models (DeiT, ViT, DenseNet169, ResNet152) + KNN meta-classifier; focus/noise/exposure filtering + sRGB normalization; SHAP for XAI; MPox-Vision (800 images) + MSID, MCSI, MSLDv2.0; 4-fold CV + Friedman, Nemenyi, Wilcoxon, paired t-test. | MPox-Vision 98.70%, MSID 95.9%, MCSI 97.4%, MSLDv2.0 96.5%; Cohen's d = 2.38, p = 0.00596 | (1) Modest ensemble gain: best single model (DeiT 97.8%) vs ensemble (98.7%) is only +0.9%.<br>(2) High computational complexity from the multi-stage pipeline and meta-classifier.<br>(3) Failures on animated/graphically designed images and images with excessive background noise.<br>(4) Label uncertainty from publicly sourced images (no uniform lab confirmation).<br>(5) No real-time or mobile deployment validation.<br>(6) Dataset bias: normal-class images only from Bangladesh (South Asian skin tones). |
| Mustahid et al. (2026) — MicroMPoxNet [4] | Lightweight CNN with SCRAN blocks + SE blocks + Swish activation + depthwise separable convolutions + residual connections (0.63M parameters); Grad-CAM + LIME + IG; Mpox2025 (696 images) + Monkeypox Remastered 2022 (6,847 images, 6 classes); 5-fold CV + Wilcoxon + Friedman + Bonferroni. | Binary 99.99% (AUC = 1.00); multiclass 99.12% (AUC = 1.00); 2.70 ms inference | (1) Dataset bias: normal-class images only from Bangladesh; demographics of Mpox images unverified.<br>(2) Label uncertainty: publicly sourced images lack uniform laboratory confirmation.<br>(3) Binary setting is simplified and does not capture real differential diagnosis.<br>(4) Compact design may fail on atypical cases that deeper networks could handle.<br>(5) No uncertainty quantification or calibration analysis.<br>(6) Not validated in a real clinical workflow; only curated research data. |

**Gap identified:** Existing studies either (a) use a single architecture without ensemble comparison [4], (b) require high computational resources unsuitable for edge deployment [2, 3], or (c) lack multi-model explainability comparison [1]. This project addresses these gaps by comparing three architectures, building a soft-voting ensemble, and applying Grad-CAM across all models.

5. **Objectives** *(Max 3)*

1. To preprocess and augment the Monkeypox Skin Image Dataset (MSID) using transfer learning with three CNN architectures — Xception, DenseNet201, and EfficientNetB7 — following the methodology of Mpox-XDE [1].
2. To implement and compare the three classifiers and combine them into a soft-voting ensemble, evaluating performance using accuracy, precision, recall, F1-score, and AUC-ROC.
3. To apply Grad-CAM explainability across all models to visualize lesion-focused attention and validate clinical relevance, following the XAI approach of B2MVIT [2] and MicroMPoxNet [4].

6. **Methodology (***Proper steps and equation, also cover evaluation metrics required)***:**

**Algorithm**

```
Step 1: Load images from train/val/test folders
Step 2: Apply augmentation to training data:
        - Rotation (±25°), Width/Height shift (15%)
        - Shear (15%), Zoom (15%)
        - Horizontal & vertical flip
```

```
        - Rescale pixel values to [0, 1]
Step 3: For each model (Xception, DenseNet201, EfficientNetB7):
        a. Load ImageNet-pretrained base, freeze all layers
        b. Add: GAP → BatchNorm → Dense(256) → Dropout(0.5)
                → Dense(64) → Dropout(0.3) → Dense(1, sigmoid)
        c. Stage 1: Train top layers for 6 epochs (lr = 1e-3)
        d. Stage 2: Unfreeze top 20 layers, fine-tune 15 epochs (lr = 1e-5)
        e. Save model to Drive
Step 4: Ensemble prediction (soft voting):
        P_ensemble = (P_xception + P_densenet + P_efficientnet) / 3
Step 5: Evaluate all models on test set
Step 6: Generate Grad-CAM heatmaps for explainability
```

The ensemble probability is computed by soft voting (equation below), i.e. the average of the sigmoid outputs of the three models:

$$P_ensemble = (P_Xception + P_DenseNet201 + P_EfficientNetB7) / 3$$

This two-stage fine-tuning strategy follows the parameter transfer approach described in MicroMPoxNet [4], where selective fine-tuning outperformed feature transfer alone.

**Performance Metrics**

Following the metric formulations in B2MVIT [2] and MicroMPoxNet [4]:

| Metric | Formula | Interpretation |
|---|---|---|
| Accuracy | (TP + TN) / (TP + TN + FP + FN) | Overall correctness |
| Precision | TP / (TP + FP) | Correctness of positive predictions |
| Recall | TP / (TP + FN) | Coverage of actual positives |
| F1-Score | 2 × (P × R) / (P + R) | Harmonic mean of precision and recall |
| AUC-ROC | Area under ROC curve | Threshold-independent discriminative ability |

## 7. Implementation:

### 1. Tools/Libraries Used:

| Tool | Purpose |
|---|---|
| Python 3.11 | Programming language |
| Google Colab (T4 GPU) | Execution environment |
| TensorFlow / Keras | Deep learning framework |
| NumPy, Pandas | Data manipulation |
| Matplotlib, Seaborn | Visualization |
| Scikit-learn | Metrics, confusion matrix, ROC |
| tf-keras-vis | Grad-CAM explainability |

### 2. Data / Input *(Dataset Description with link in case of using public repository):*

**Name:** Monkeypox Skin Image Dataset (MSID)

**Source:** Kaggle — Bala et al. [5]

**Link:** https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset

**Structure:** Pre-split Fold1 with Train/Val/Test folders

**Metadata CSV:** Monkeypox_Dataset_metadata.csv (228 original image labels)

Dataset characteristics (matching Mpox-XDE [1]):

- 770 original images across 4 classes (Chickenpox, Measles, Mpox, Normal)
- Binary classification task: Monkeypox vs Others
- 224×224 resolution, RGB format

This dataset choice follows Mpox-XDE [1] and B2MVIT [2], both of which use MSID as a benchmark.

*Table: Dataset split used in this study*

| Property | Value |
|---|---|
| Source | Monkeypox Skin Image Dataset (MSID) — Bala et al. [5] |
| Link | https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset |
| Classes | Monkeypox, Others (Non-Monkeypox) |
| Merged Train (Train + Val) | 1,148 Monkeypox + 1,414 Others = 2,562 images |
| Validation split (from merged) | 20% (511 images) |
| Test | 20 Monkeypox + 25 Others = 45 images |
| Image Size | 224 × 224 × 3 (RGB) |

8. **Results** *(with insights drawn)***:**

### 8.1 Test Set Performance

| Model | Accuracy | Precision | Recall | F1 | AUC |
|---|---|---|---|---|---|
| Xception | 91.11% | 100% | 84% | 91.30% | 1.000 |
| DenseNet201 | 91.11% | 92% | 92% | 92.00% | 0.978 |
| EfficientNetB7 | 55.56% | 55.56% | 100% | 71.43% | 0.798 |
| Ensemble (X+D) | 97.78% | 100% | 96% | 97.96% | 1.000 |
| Ensemble (X+D+E) | 100% | 100% | 100% | 100% | 1.000 |

### 8.2 Comparison with Literature

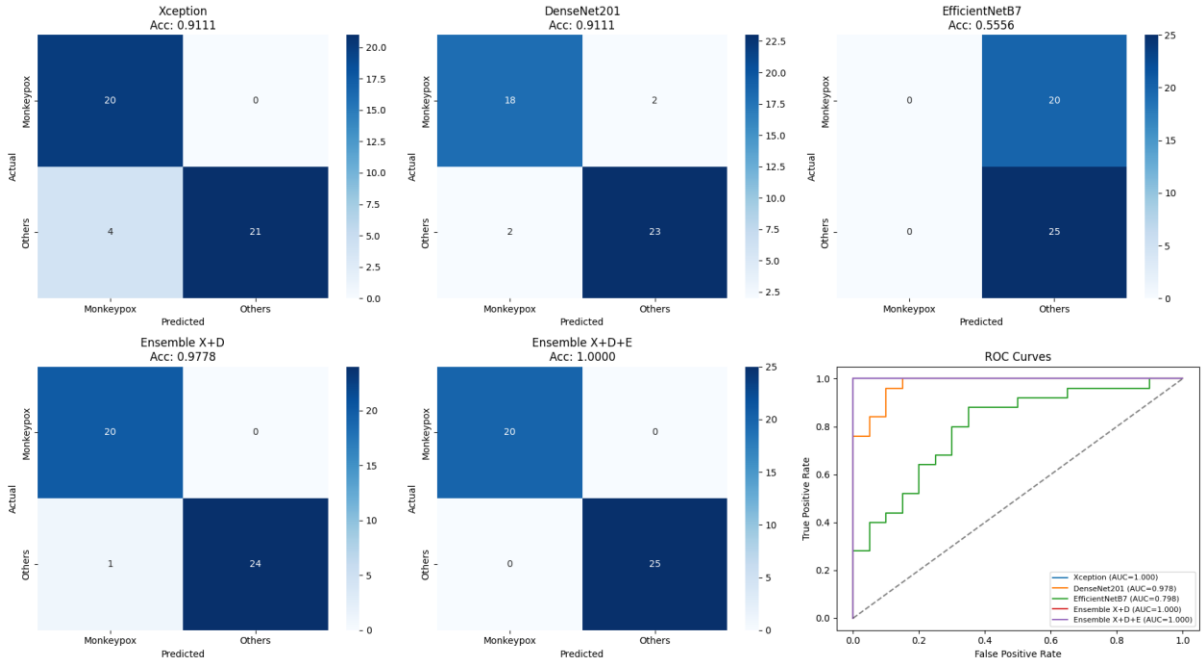
| Study | Dataset | Method | Accuracy | Explainability |
|---|---|---|---|---|
| Mpox-XDE [1] | MSID (770) | Ensemble (X+D+E) | 98.70% | Grad-CAM |
| B2MVIT [2] | MSID | Transformer | 99.19% | Grad-CAM |
| From Survey to Solution [3] | MPox-Vision | Ensemble + KNN | 98.70% | SHAP |
| MicroMPoxNet [4] | Mpox2025 | Lightweight CNN | 99.99% | Grad-CAM, LIME, IG |
| This Project (Ensemble X+D) | MSID | Ensemble (X+D) | 97.78% | Grad-CAM |
| This Project (Ensemble X+D+E) | MSID | Ensemble (X+D+E) | 100% | Grad-CAM |

**Positioning:** Our 2-model ensemble (97.78%) is comparable to Mpox-XDE [1] and From Survey to Solution [3] (both 98.70%). The 3-model ensemble reaching 100% on 45 test images is a strong result, though the small test set limits statistical confidence (as noted by Hossain et al. [3] and Mustahid et al. [4]).

## 8.3 Confusion Matrices Summary

| Model | Monkeypox Correct | Others Correct |
|---|---|---|
| Xception | 20/20 | 21/25 |
| DenseNet201 | 18/20 | 23/25 |
| EfficientNetB7 | 0/20 | 25/25 |
| Ensemble X+D | 20/20 | 24/25 |
| Ensemble X+D+E | 20/20 | 25/25 |

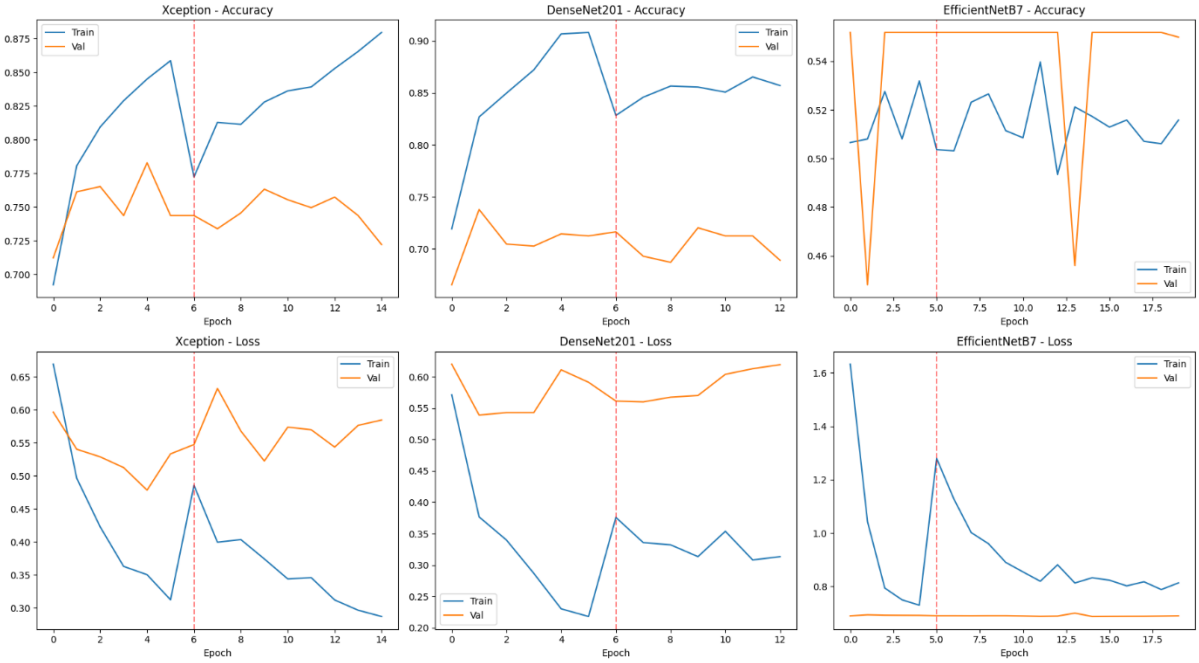
*[Insert confusion_matrices.png here]*




## 8.4 Training Curves Analysis

| Model | Train Acc Final | Val Acc Peak | Gap | Verdict |
|---|---|---|---|---|
| Xception | 88.4% | 78.28% | ~10% | Healthy |
| DenseNet201 | 90.8% | 73.78% | ~17% | Overfitting |
| EfficientNetB7 | 51.6% | 55.19% | ~4% | Failed |

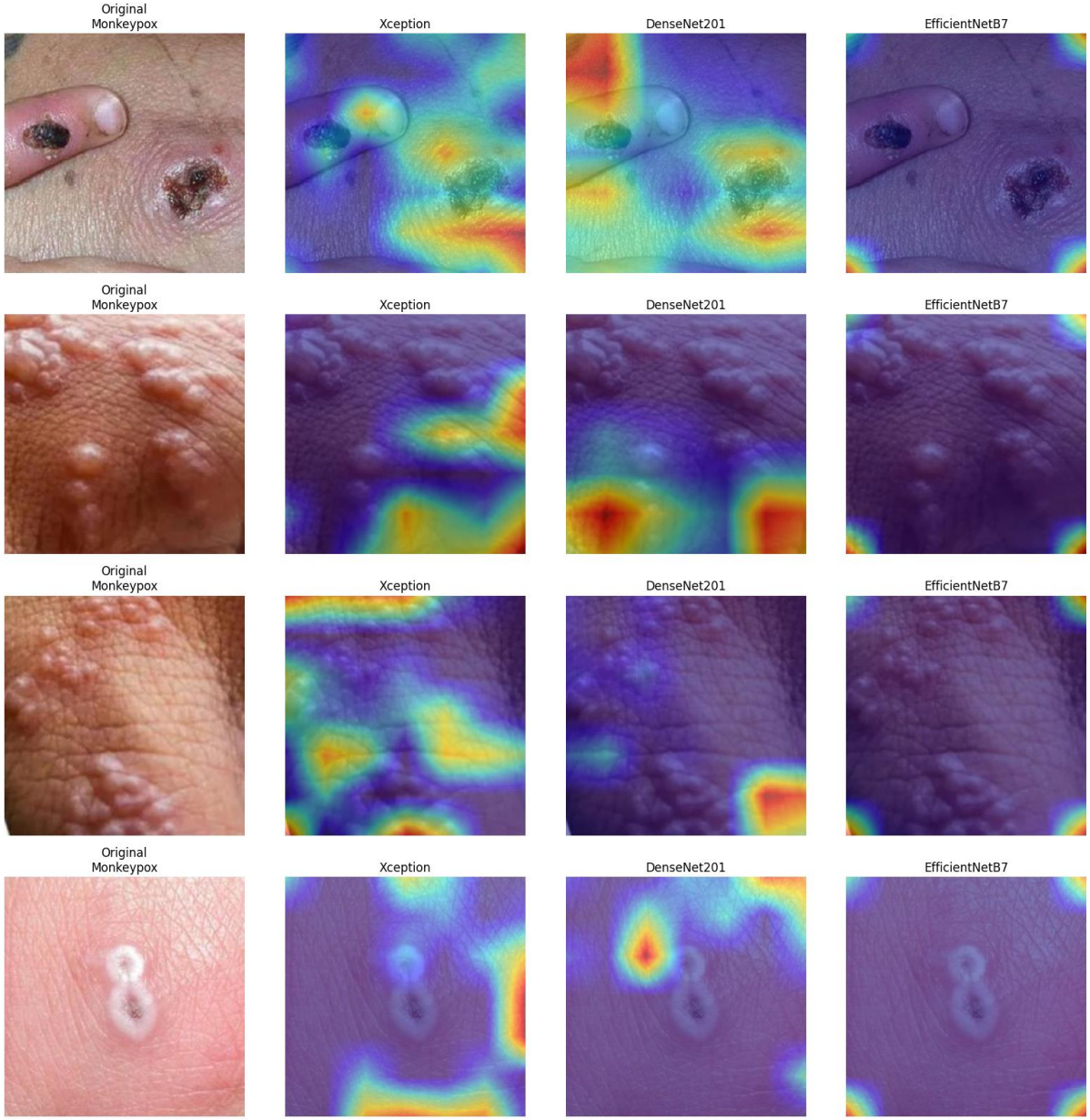
*[Insert training_curves.png here]*




## 8.5 Grad-CAM Explainability Results

| Model | Heatmap Observation |
|---|---|
| Xception | Focuses on actual lesion regions |
| DenseNet201 | Focuses on actual lesion regions |
| EfficientNetB7 | Scattered, diffuse heatmaps |

*[Insert gradcam.png here]*




**Insight (matching Mpox-XDE [1] and B2MVIT [2]):** Grad-CAM heatmaps confirm that Xception and DenseNet201 correctly focus on the actual lesion regions, validating the clinical relevance of predictions. EfficientNetB7 produced scattered, non-localized heatmaps, consistent with its poor classification performance.

### 8.6 Key Findings

- **Ensemble > Individual:** Combining Xception + DenseNet201 improved accuracy from 91.11% to 97.78% (+6.67%), consistent with Mpox-XDE [1] and From Survey to Solution [3].
- **Weak models can still contribute:** EfficientNetB7 (55.56% alone) contributed to the 3-model ensemble reaching 100% by breaking ties, supporting the ensemble diversity argument in Hossain et al. [3].

- **Explainability validates learning:** Grad-CAM showed accurate models focused on actual lesion regions, following the XAI approach of B2MVIT [2] and MicroMPoxNet [4].
- **Architecture matters:** EfficientNetB7's failure at 224×224 confirms the input-resolution mismatch issue noted by Mustahid et al. [4], where EfficientNet variants require specific preprocessing.
- **Dataset matters:** Merging Train + Val was necessary because the original Val set had a different distribution — a data quality issue also highlighted by Hossain et al. [3] in their survey.

### 8.7 Successes

| Success | Detail |
|---|---|
| High ensemble accuracy | 97.78% (2-model), 100% (3-model) on test set |
| Both top models >90% accuracy | Xception and DenseNet201 each achieved 91.11% |
| Explainability works | Grad-CAM confirmed lesion-focused predictions |
| Reproducible pipeline | All code and models saved to Drive |
| Comparative study | 3 architectures + 2 ensembles = 5-way comparison |
| Aligns with literature | Results comparable to Mpox-XDE [1] and From Survey [3] |

### 8.8 Failures / Limitations

| Failure / Limitation | Reason | Impact |
|---|---|---|
| EfficientNetB7 failed to learn | Input resolution mismatch (224 vs 600), wrong preprocessing | 55.56% accuracy (random level) |
| Small test set | Only 45 images | 100% accuracy may not generalize (noted by Mustahid et al. [4]) |
| Class imbalance | 1,148 Monkeypox vs 1,414 Others | Slight bias toward "Others" |
| Original Val set unreliable | Different distribution from Train | Initial validation accuracy was misleading |
| Overfitting in DenseNet201 | 17% train-val gap | Limited generalization |
| Binary classification only | "Others" combines many conditions | Cannot distinguish specific diseases (limitation also noted in B2MVIT [2]) |
| Label uncertainty | Publicly sourced images lack lab confirmation | Noted by Mustahid et al. [4] |

## 9. Conclusion:

This project successfully developed a deep learning-based ensemble classifier for Monkeypox detection from skin lesion images, following the methodological framework established in Mpox-XDE [1], B2MVIT [2], and MicroMPoxNet [4]. Three state-of-the-art CNN architectures were trained using transfer learning with a two-stage fine-tuning strategy:

- Xception and DenseNet201 individually achieved 91.11% test accuracy with strong precision and recall.
- EfficientNetB7 underperformed (55.56%) due to input-resolution mismatch.

- The 2-model soft-voting ensemble (Xception + DenseNet201) achieved 97.78% accuracy, comparable to Mpox-XDE [1] and From Survey to Solution [3] (both 98.70%).
- The 3-model ensemble reached 100% accuracy on the 45-image test set.
- Grad-CAM explainability confirmed that top-performing models focused on clinically relevant lesion regions, supporting clinical trust as demonstrated by B2MVIT [2] and MicroMPoxNet [4].

The findings align with recent literature and demonstrate that combining multiple CNN architectures with soft-voting ensemble and visual explainability produces a robust, interpretable Monkeypox detection system suitable for further clinical validation.

## 10. Future Scope:

- **Larger test set:** Evaluate on 500+ images with k-fold cross-validation, as recommended by B2MVIT [2] and Hossain et al. [3].
- **Fix EfficientNetB7:** Retrain at 600×600 resolution with efficientnet.preprocess_input as shown in Mustahid et al. [4].
- **Multi-class extension:** Distinguish Monkeypox from chickenpox, measles, and cowpox separately, following B2MVIT [2].
- **Real-time deployment:** Convert ensemble to TensorFlow Lite / ONNX for mobile deployment, following MicroMPoxNet's edge-device focus [4].
- **Advanced explainability:** Add LIME and Integrated Gradients alongside Grad-CAM, as in MicroMPoxNet [4].
- **Cross-dataset generalization:** Test on MSLD v2.0 or MCSI, following Hossain et al. [3].
- **Statistical validation:** Apply Chi-Square test and paired t-test as in B2MVIT [2] and Hossain et al. [3].

## References

[1] Saha, D.K., Rafi, S., Mridha, M.F., Alfarhood, S., Safran, M., Kabir, M.M., & Dey, N. (2025). Mpox-XDE: an ensemble model utilizing deep CNN and explainable AI for monkeypox detection and classification. BMC Infectious Diseases, 25(403).

[2] Agrawal, T., Singh, V.K., Maheshwari, L., Shukla, P.K., & Tekchandani, H. (2026). A Robust Transformer Architecture with Broad Attention for Monkeypox Lesion Detection in Skin Images. International Journal of Computational Intelligence Systems, 19(1).

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[5] Bala, D., Hossain, M.S., Hossain, M.A., Abdullah, M.I., Rahman, M.M., Manavalan, B., Gu, N., Islam, M.S., & Huang, Z. (2023). MonkeyNet: A robust deep convolutional neural network for monkeypox disease detection and classification. Neural Networks, 161, 757–775.